Crisis Regime Analysis

Onset / Chronic / Recovery / Stable — Fact-Based Results

1. Regime Definitions

The regime of each test observation is determined by combining the current outcome (crisis or not) with whether the same district was already in crisis the prior period (ipc_lag_1). This is a post-hoc diagnostic label — it is not used during training, only for interpreting results.

- Onset — crisis now, no crisis prior: a district tipping into crisis for the first time
- Chronic — crisis now, crisis also prior: a district locked in persistent crisis
- Recovery — no crisis now, crisis prior: a district that just exited crisis
- Stable — no crisis now, no crisis prior: a district with no recent crisis history

2. Test-Set Composition (2-year window, 7 folds, n=2,148)

Regime	Obs	Crisis obs	% of crises	Prevalence
Chronic	460	460	72.3%	100%
Onset	176	176	27.7%	100%
Recovery	98	0	—	0%
Stable	1,414	0	—	0%
Total	**2,148**	**636**	**100%**	**29.6%**

3. Chronic — History Is Enough

n = 460 crisis observations (72.3% of all crises). AR-Only recall: 97% AR+News recall: 96%.
AR-Only median P(crisis): 0.961 AR+News median: 0.972

These are districts already in crisis last period. The model knows this through ipc_lag_1, which alone pushes the predicted probability well above 0.5 — no news is needed. Both models detect every chronic crisis in every fold. Adding news information actually lowers the predicted probability for 52.6% of chronic cases (median shift -0.001), because news coverage of a chronic crisis looks different from news coverage of an emerging one. The AR margin is wide enough that this makes no difference to detection — but it illustrates the models are using different signals.

4. Onset — Where News Earns Its Keep

n = 176 crisis observations (27.7% of all crises). AR-Only recall: 11.4% (20 of 176) AR+News recall: 14.2% (25 of 176)
AR-Only median P(crisis): 0.061 AR+News median: 0.151

These are the hardest cases — a district falling into crisis for the first time, with no prior signal in the data. The AR model, having nothing in the history to flag, assigns a very low probability (90th percentile only 0.583). News fills the gap: reports of conflict, food insecurity, or displacement arrive before the IPC assessment does, lifting the median predicted probability from 0.061 to 0.151. This is enough to push 25 of 176 onset crises over the detection threshold — though 86% remain undetected even with news.

Fold-by-fold onset breakdown:

Fold	n onset	AR recall	AR+News recall	Net saves
1	11	9%	73%	+7
2	2	100%	0%	-2
3	5	20%	0%	-1
4	26	8%	0%	-2
5	43	2%	9%	+3
6	53	23%	23%	+0
7	36	3%	3%	+0

Net onset saves: 0 total. News improves onset detection in 2 of 7 folds; in the other 5, neither model detects a single onset crisis — the news signal was not strong enough.

Crisis Regime Analysis (continued)

5. Recovery — The AR Model Cannot Let Go

n = 98 observations, all non-crisis (target=0). AR-Only false alarm rate: 100% AR+News false alarm rate: 87%.
AR-Only median P(crisis): 0.921 AR+News median: 0.863

Recovery districts have just exited crisis — they are not in crisis now, but they were last period. The AR model sees ipc_lag_1=1 and treats them identically to chronic crisis districts, assigning crisis-level confidence to all 98 of them (median 0.921).
It has no way to distinguish 'was in crisis and still is' from 'was in crisis and recovered'.
Adding news pulls the predicted probability down (median 0.921 → 0.863),
suggesting the news landscape of a recovering district does look different — but not different enough to cross back below 0.5. Both models flag every recovery district as a false alarm.

6. Stable — News Is Sensitive to Quiet Districts

n = 1,414 observations, all non-crisis (target=0). AR-Only false alarm rate: 6% AR+News false alarm rate: 4.3% (61 cases).

With no prior crisis signal, the AR model is very conservative — it assigns low probabilities to all stable districts and produces zero false alarms. News changes this. The combined model raises the predicted probability for 86% of stable observations (median Δ = +0.200), the largest and most consistent upward shift across all four regimes. This makes intuitive sense: news coverage of conflict, weather events, or food stress can be elevated in a district that has not yet reached IPC Phase 3. In 1,353 of 1,414 cases the model
stays correctly below 0.5, but in 61 it crosses the threshold — generating false alarms in places where the IPC record shows no crisis, current or prior.

7. Overall False Alarm Rates (all non-crisis observations)

Total non-crisis observations: 1,512 (stable 1,414 + recovery 98)
AR-Only: 185 false alarms (12.2%) — all from recovery districts
AR+News: 146 false alarms (9.7%) — recovery (98) + stable (61)

AR-Only false alarms come entirely from its inability to recognise recovery — a structural limitation of the lag feature. AR+News inherits that problem and adds a new one: it raises probabilities in stable districts enough to trigger 61 additional false alarms.
The net result is -39 more false alarms than AR-Only, a trade-off against the 0 additional onset crises it correctly detects.

8. Net Balance

AR+News uniquely detects 23 crisis cases that AR-Only misses.
AR+News loses 23 detection(s) that AR-Only makes — net regression mainly in chronic.
Net saves: +0 (0.0% of 636 crisis observations). Both detect: 442. Neither detects: 148 (23.3%).

The intuitive story across regimes is consistent. For chronic crises, past history is the dominant signal and news is redundant. For onset crises, history is silent and news becomes the only available signal — imperfect, but real. For recovery districts, the lag feature misfires because it cannot distinguish exit from persistence; news nudges the model in the right direction but not far enough. For stable districts, news is sensitive to early-warning signals that may not yet be visible in IPC assessments — a potential strength, but one that also generates false alarms where no crisis materialises.